

Database Management System

EXPERIMENT 11 CREATING VIEWS

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1. Create EMPLOYEE_VU view based on employee numbers, names, and department numbers

```
CREATE OR REPLACE VIEW employee_vu AS
SELECT employee_id, first_name, last_name AS employee,
department_id
FROM employees;
```

Expected Output:

View created.

2. Display the contents of the EMPLOYEE_VU view

```
SELECT * FROM employee_vu;
```

Expected Output:

EMPLOYEE_ID	FIRST_NAME	EMPLOYEE	DEPARTMENT_ID

```

-----
          100 Steven                King
90
          101 Neena                 Kochhar
90
          102 Lex                   De Haan
0
          103 Alexander             Hunold
60
          104 Bruce                 Ernst
60
          105 David                 Austin
60
          106 Valli                 Pataballa
60
          107 Diana                 Lorentz
60
          108 Nancy                 Greenberg
100
          109 Daniel                Faviet
100
          110 John                  Chen
100
          ... (continues for all employees)

```

3. Select view name and text from USER_VIEWS

```

SELECT view_name, text
FROM user_views
WHERE view_name = 'EMPLOYEE_VU';

```

Expected Output:

VIEW_NAME	TEXT
EMPLOYEE_VU	SELECT employee_id, first_name, last_name AS employee, department_id FROM employees

4. Using EMPLOYEE_VU view, display all employee names and departments

```

SELECT employee, department_id
FROM employee_vu
ORDER BY department_id;

```

Expected Output:

EMPLOYEE	DEPARTMENT_ID
Whalen	10
Fay	20
Hartstein	20
Marvis	30
Baida	30
Colmenares	30
Himuro	30
Khoo	30
Raphaely	30
Tobias	30
... (continues grouped by department)	

5. Create DEPT50 view with restrictions

```

CREATE OR REPLACE VIEW dept50 AS
SELECT employee_id AS empno, last_name AS employee,

```

```

department_id AS deptno
FROM employees
WHERE department_id = 50
WITH CHECK OPTION CONSTRAINT dept50_ck;

```

Expected Output:

View created.

6. Display structure and contents of DEPT50 view

```
DESC dept50;
```

Expected Output:

Name	Null?	Type
EMPNO		NUMBER(6)
EMPLOYEE		VARCHAR2(25)
DEPTNO		NUMBER(4)

```
SELECT * FROM dept50;
```

Expected Output:

EMPNO	EMPLOYEE	DEPTNO
120	Weiss	50
121	Fripp	50
122	Kaufling	50
123	Vollman	50
124	Mourgos	50
125	Nayer	50
126	Mikkilineni	50

127 Landry	50
128 Markle	50
129 Bissot	50
130 Atkinson	50
... (all employees from department 50)	

7. Attempt to reassign Matos to department 80

```
UPDATE dept50
SET deptno = 80
WHERE employee = 'Matos';
```

Expected Output:

```
ERROR at line 1:
ORA-01402: view WITH CHECK OPTION where-clause violation
```

8. Create SALARY_VU view with employee details, department names, salaries, and grades

```
CREATE OR REPLACE VIEW salary_vu AS
SELECT e.last_name AS "Employee",
       d.department_name AS "Department",
       e.salary AS "Salary",
       jg.grade_level AS "Grade"
FROM employees e
JOIN departments d ON e.department_id = d.department_id
JOIN job_grades jg ON e.salary BETWEEN jg.lowest_sal AND
jg.highest_sal;
```

Expected Output:

View created.

Display SALARY_VU view contents:

```
SELECT * FROM salary_vu ORDER BY "Salary" DESC;
```

Expected Output:

Employee	Department	Salary
G		

-		
King	Executive	24000
F		
Kochhar	Executive	17000
E		
De Haan	Executive	17000
E		
Hartstein	Marketing	13000
D		
Russell	Sales	14000
D		
Partners	Sales	13500
D		
Errazuriz	Sales	12000
D		
Higgins	Accounting	12000
D		
... (continues with all employees)		

Additional View Examples:

Simple View with Column Aliases

```
CREATE OR REPLACE VIEW emp_simple AS
SELECT employee_id AS id,
       first_name || ' ' || last_name AS full_name,
       salary,
       department_id
FROM employees;
```

Complex View with Aggregation

```
CREATE OR REPLACE VIEW dept_summary AS
SELECT d.department_name,
       COUNT(e.employee_id) AS employee_count,
       AVG(e.salary) AS avg_salary,
       MAX(e.salary) AS max_salary,
       MIN(e.salary) AS min_salary
FROM departments d
LEFT JOIN employees e ON d.department_id = e.department_id
GROUP BY d.department_name;
```

Read-Only View

```
CREATE OR REPLACE VIEW emp_readonly AS
SELECT employee_id, first_name, last_name, hire_date
FROM employees
WITH READ ONLY;
```

Key View Concepts Demonstrated:

1. **Simple Views** - Based on single table with no functions
2. **Complex Views** - Contain joins, groups, or functions
3. **Column Aliases** - Renaming columns in views
4. **WITH CHECK OPTION** - Prevents DML that would exclude rows from view

5. **WITH READ ONLY** - Prevents all DML operations on view
6. **Data Dictionary** - USER_VIEWS contains view definitions
7. **View Updating** - Some views allow DML, others don't